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A continental record of the late Campanian negative carbon isotopic excursion (LCE, late Campanian event) in the Dinosaur Park Formation, Alberta, Canada

Muditha Goonetilleke *, Ricardo L. Silva, Paul R. Durkin

PaleoSed+ Research Group, Department of Earth Sciences, Clayton H. Riddell Faculty of Environment, Earth, and Resources, University of Manitoba, 125 Dysart Road, Winnipeg, Manitoba R3T 2N2, Canada

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ABSTRACT

The Campanian was the longest stage of the Late Cretaceous and witnessed multiple global carbon cycle disruption episodes, including the negative carbon isotope excursion (CIE) known as the Late Campanian Event (LCE). However, Campanian-aged $\delta^{13}C$ perturbations are primarily documented in marine environments, raising questions about their record in terrestrial settings and the potential use of terrestrial stable carbon isotope records from the Campanian as a chemostratigraphic tool. In this study, we present a stable carbon isotope record from fossil organic matter preserved in the alluvial floodplains of the Dinosaur Park Formation, which is radiometrically dated to the time interval containing the LCE. Results reveal a ~0.8 ‰ negative CIE recorded in floodplain fossil organic matter. We show that the negative CIE is not a result of changing organic matter or diagenesis and that it likely reflects the global atmospheric $\delta^{13}C$ signal during the Campanian. We interpreted the identified negative CIE as a continental record of the LCE. When we couple our $\delta^{13}C_{TOC}$ record with the Dinosaur Park Formation's age model, we can correlate our section with other well-dated European marine stratigraphic sections. Our findings demonstrate that Campanian carbon cycle perturbations are preserved in terrestrial environments. For the first time, we recognize that the deposition of the world-famous dinosaur-bearing Dinosaur Park Formation in Alberta was contemporaneous with a major global carbon cycle perturbation episode. Our findings contribute to contextualizing ongoing research and re-evaluating previous studies on dinosaur evolution and other contemporary groups, within the framework of planetary-scale Earth system disturbances.

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1. Introduction

Surficial crustal carbon (C) reservoirs, such as marine and terrestrial environments, continuously exchange carbon with the atmosphere, and disruption to the carbon fluxes between them are often recorded synchronously across extensive spatial scales (Schubert and Jahren, 2013). Perturbations to the global carbon cycle are often expressed as carbon isotope excursions (CIEs) in the global $\delta^{13}C$ geological record; thus, determining $\delta^{13}C$ in a wide range of substrates in a given stratigraphic section may offer valuable insights into internal geodynamic and surficial processes operating at that time (Weissert, 1989; Berner, 1999, 1998; Kump and Arthur, 1999; Weissert et al., 2008; Jenkyns, 2010; Rodrigues et al., 2021, 2019; Ruhl et al., 2022). When combined with other time markers, such as radiometric ages and index fossils, CIEs facilitate the correlation of local records with the global $\delta^{13}C$ signal (Weissert, 1989; Weissert et al., 2008, 1998; Gröcke, 2020). However, environmental, biological, and diagenetic bias in $\delta^{13}C$ records may obscure the record of global carbon cycle processes, complicating the correlation of $\delta^{13}C$ records across broader geographical areas (e.g., Tyson, 1995; Silva et al., 2015, 2013, 2011; Fang et al., 2013; Silva and Duarte, 2015).

The Campanian (72.2–83.6 Ma), the longest duration stage of the Late Cretaceous, witnessed multiple carbon cycle perturbations (Jarvis et al., 2002; Cramer and Jarvis, 2020). Many of these perturbations have been linked to sea-level changes (Jarvis et al., 2002), bottom-water oxygenation in oceanic environments (Jung

* Corresponding author. Clayton H. Riddell Faculty of Environment, Earth, and Resources, University of Manitoba, 125 Dysart Road, Winnipeg, Manitoba R3T 2N2, Canada.
E-mail address: goonetim@myumanitoba.ca (M. Goonetilleke).

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et al., 2013; Linnert et al., 2018), and intense weathering and erosion in terrestrial settings (Chenot et al., 2018). These processes ultimately caused excess carbon release or burial within surficial reservoirs, which, in a steady-state carbon cycle (Berner, 2006, 1999, 1998; Kump and Arthur, 1999), are commonly recorded as negative (i.e. release of carbon) or positive (i.e. burial of carbon) CIEs. While Campanian-aged CIEs are primarily documented and correlated in the marine realm, it remains uncertain whether and where similar $\delta^{13}C$ trends are preserved in Campanian-aged terrestrial records and if they can be used as a chemostratigraphic tool to correlate terrestrial and marine sedimentary sections globally.

Based on radiometric ages from Ramezani et al. (2022), the upper Campanian alluvial deposits of the Dinosaur Park Formation in Canada fall within the time interval containing the Late Campanian Event (LCE), a ~0.5–1 ‰ negative CIE defined in numerous previous studies in marine and marginal marine sections (e.g., Jarvis et al., 2002; Linnert et al., 2018; Pearce et al., 2022). We hypothesize that the organic components (i.e., kerogen) preserved in the Dinosaur Park Formation sedimentary rocks record contemporary changes in global-scale carbon cycling processes. The objective of our study is to obtain a $\delta^{13}C$ curve from organic matter preserved in the continuously accumulating floodplains of the Dinosaur Park Formation and compare it with other global stratigraphic sections. We also address the uncertainty of whether $\delta^{13}C$ from the studied Campanian-aged fully terrestrial record can be used as a chemostratigraphic tool to correlate sedimentary sections globally, while evaluating the extent to which terrestrial records reflect changes in global-scale carbon cycling dynamics (i.e., the LCE), and how these might have affected terrestrial biological ecosystems.

2. Geological setting

The Dinosaur Park Formation (DPF), part of the Upper Cretaceous Belly River Group (Fig. 1A), records a meandering fluvial system developed on a broad alluvial and coastal plain (Eberth, 2005). The lower to middle stratigraphic units of the Dinosaur Park Formation are characterized by channel-belt deposits composed of sandstones and mudstones, along with overbank deposits consisting primarily of mudstones and shales (Eberth, 2005). The upper unit of the Dinosaur Park Formation, which contains the Lethbridge Coal Zone (LCZ), consists predominantly of carbonaceous mudstone, depicting a waterlogged coastal swamp,

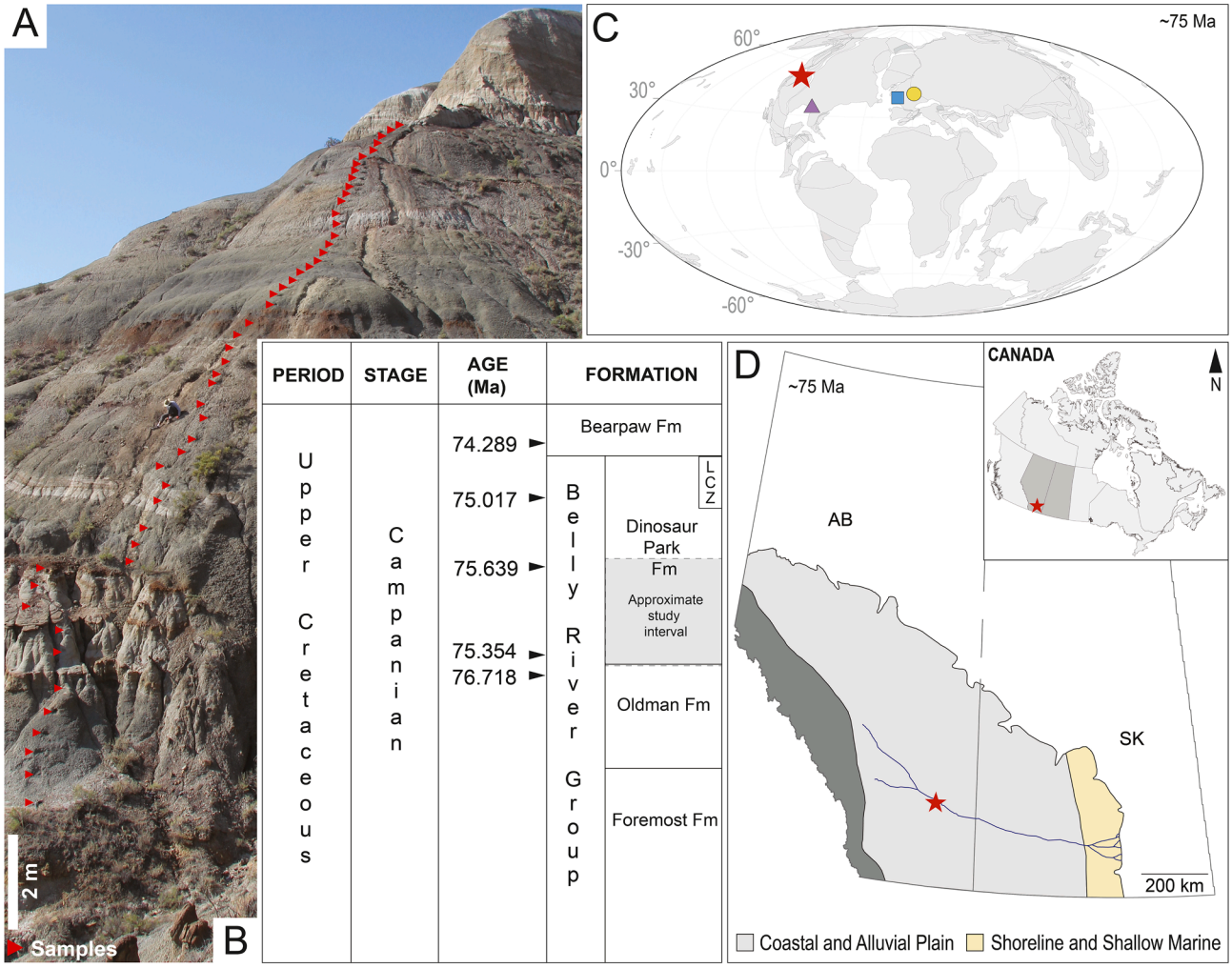


Fig. 1. (A) Measured stratigraphic section and sample collection sites (red arrows). Person for scale (modified from Goonetilleke et al., 2025). (B) Stratigraphy and chronological ages of the Belly River Group in Alberta alongside the approximate stratigraphic positioning of the study interval (modified from Eberth et al., 2023). Fm - Formation, LCZ - Lethbridge Coal Zone. (C) Approximate Campanian paleogeographic positions of Dinosaur Provincial Park (red star), a North American section from the USA (purple triangle), and Boreal Sea sections in northern Germany (yellow dot) and eastern UK (blue square), used for correlation (map generated using odsn.de, 2014). (D) Paleogeographic reconstruction of the Dinosaur Park Formation (modified from Mayo et al., 2023), with Dinosaur Provincial Park marked by a red star. AB - Alberta, SK - Saskatchewan.

which is overlain by marine shales of the Bearpaw Formation (Eberth, 2005, 1996) (Fig. 1B). The poorly cemented sandstones of the Dinosaur Park Formation, appearing grayish-brown to light gray and rounded at the surface, are clearly distinguishable from the underlying Oldman Formation, a braided fluvial system, which consists of well-cemented, light-colored sandstones that appear blocky and smooth when exposed (Eberth, 2005). The Oldman Formation-Dinosaur Park Formation contact can be traced regionally throughout exposures (Eberth and Hamblin, 1993). Bentonite deposits within both the Belly River Group and the Bearpaw Formation have been radiometrically dated using U–Pb geochronology by Ramezani et al. (2022), contributing to a comprehensive age model for the Dinosaur Park Formation (Eberth et al., 2023) (Fig. 1B). The model brackets the Dinosaur Park Formation between 76.470 + 0.140/-0.084 (2σ internal uncertainties) Ma and 74.44 + 0.30/-0.11 (2σ internal uncertainties) Ma, with the lower, middle, and upper successions of the formation having accumulation rates of 4.90 ± 0.43, 4.11 ± 0.21, and 2.99 ± 0.10 cm/kyr, respectively (Eberth et al., 2023).

2.1. Studied section

A ~30 m thick sedimentary succession from the northwestern sector (Steveville, 50.832°N, 111.590°W) of Dinosaur Provincial Park (Alberta, Canada) was selected for this study (Fig. 1A and C). Recent work demonstrates that the studied section from the Dinosaur Park Formation preserves a fluvial succession, unaffected by tidal processes of the Western Interior Seaway (Durkin et al., 2020; Mayo et al., 2023; Goonetilleke et al., 2025) (Fig. 1D).

Detailed palynofacies and sedimentological analysis for the section are presented in Goonetilleke et al. (2025), where the study demonstrates that the section comprises five sub-environments with distinct kerogen (fossil organic matter) assemblages. These sub-environments can be broadly categorized as channel-dominated floodplains (Fig. 2A), characterized by abundant opaque phytoclasts (Fig. 2B), and stable floodplain deposits (Fig. 2A), dominated by cuticles, membranes and palynomorphs (Fig. 2B). The channel-dominated interval extends up to ~9 m from the base, followed by the stable floodplain interval up to 31.5 m (Fig. 2A).

The Oldman Formation-Dinosaur Park Formation contact is located ~2.5 m above the base of the studied section, and a bentonite layer (Steveville Bentonite, in Thomas et al., 1990) near the top (30.5 m) feature as age constraints and stratigraphic markers for the section (Fig. 2A; Goonetilleke et al. (2025)). The bentonite correlates with the 7/29/86/1 bentonite from the same study area (Thomas et al., 1990) and the Plateau Tuff described in Eberth et al. (2023) (Fig. S1, Supplemental Material). The modelled age of the Oldman-Dinosaur Park Formation contact (76.470 + 0.140/-0.084 (2σ internal uncertainties) Ma) and the radiometric age of the Plateau Tuff (75.639 ± 0.025 (2σ internal uncertainties) Ma) brackets deposition in the studied section to an interval of about 840 kyr.

3. Methods

To extract a $\delta^{13}C$ record from the studied section, sixty-five (65) samples were collected at a 0.5 m interval. Before measuring the stable carbon isotope ratio of preserved organic matter ($\delta^{13}C_{TOC}$), 2–3 g of each sample was powdered, decarbonated in 1 M HCl for 48 h (including heating at 80 °C for 2 h to ensure acid reaction with refractory carbonate phases such as siderite and dolomite), rinsed twice with deionized water, dried, and weighed (5–10 mg) into 8 × 5 mm tin capsules for $\delta^{13}C_{TOC}$ analysis, following a methodology similar to that of Silva and Alsen (2024). The prepared samples were analyzed for $\delta^{13}C_{TOC}$ using a Costech Elemental Combustion System (ECS) 4010- Elemental Analyzer coupled to a Thermo Scientific DELTA V Plus- Isotope Ratio Mass Spectrometer (IRMS) at the Manitoba Isotope Research Facility (MIRF), University of Manitoba, Canada. For quality control of the decarbonated samples, B2151 (sediment) and B2153 (soil) were used, while international standards (USGS40, USGS41) were used at the beginning, middle, and end of each run for internal calibration. All carbon isotope ratios ($\delta^{13}C_{TOC}$, ‰) are reported relative to the Vienna PDB standard (Coplen, 1994). To visualize long-term, secular, variation in $\delta^{13}C_{TOC}$, LOESS (locally estimated scatterplot smoothing) modelling was performed using the Past 4.16c software with a bootstrap smoothing factor of 0.2.

4. Results

The $\delta^{13}C_{TOC}$ values for the measured section vary from −23.8 to −26.7 ‰ (Fig. 2C; Table S1 of the Supplemental Material). LOESS modelling shows that most $\delta^{13}C_{TOC}$ values fluctuate around an average of −25.1 ± 0.2 (2σ internal uncertainties) ‰ until ~18 m, followed by a sharp negative shift of about 0.8 ‰ within the floodplain-dominated section (Fig. 2C).

5. Discussion: the $\delta^{13}C_{TOC}$ record in the Dinosaur Park Formation

5.1. Channels vs. floodplains (time resolution)

It is estimated that deposition in the studied section took place in an interval of about 840 kyr (Fig. 3) (Eberth et al., 2023). However, channel and floodplain deposits represent differing accumulation rates (Kraus and Aslan, 1999; Miall, 2006), thus time is inconsistently distributed throughout the section. Addressing the different rates of deposition at different parts of the studied section is important when interpreting $\delta^{13}C$ records, especially when sections are sampled at a consistent interval (Gröcke, 2020).

We estimate an accumulation rate of 0.03 m/kyr for the floodplain-dominated portion (Fig. S2), while Miall (2015) indicated that channel belts accumulate at an average rate of 1 m/kyr (Fig. S2). The order-of-magnitude difference in accumulation rates between channel belts and floodplains suggests that our floodplain-dominated section preserves $\delta^{13}C$ records at time scales of 100 kyr, whereas the channel-dominated section would preserve $\delta^{13}C$ records at time scales of 10 kyr (see Fig. S2 and Supplemental Text S1). Considering the differing temporal resolutions of these deposits and our consistent sampling interval for channels and floodplain deposits, the lower channel-dominated interval and the overlying stable floodplains will be discussed separately.

5.2. Source and meaning of $\delta^{13}C_{TOC}$ record

The $\delta^{13}C$ content of sedimentary organic compounds is primarily influenced by: (1) variation in carbon sources (e.g., atmospheric CO_2 vs. oceanic dissolved inorganic carbon); (2) processes of carbon assimilation (e.g., autotrophic vs. heterotrophic); (3) differences in isotope fractionation during metabolism (e.g., C_3 vs. C_4 plants) and biosynthesis (e.g., lipids vs. carbohydrates, leaves vs. wood); and (4) cellular carbon budgets (Hayes, 1993; each comparison (vs.) reflects a difference in $\delta^{13}C$ values). This complex interplay of factors, along with diagenesis, affects the primary $\delta^{13}C_{TOC}$ signal, complicating the interpretation of absolute $\delta^{13}C_{TOC}$ values. Despite such complexities, systematically understanding the meaning of the $\delta^{13}C_{TOC}$ signal can help decipher $\delta^{13}C_{TOC}$ trends

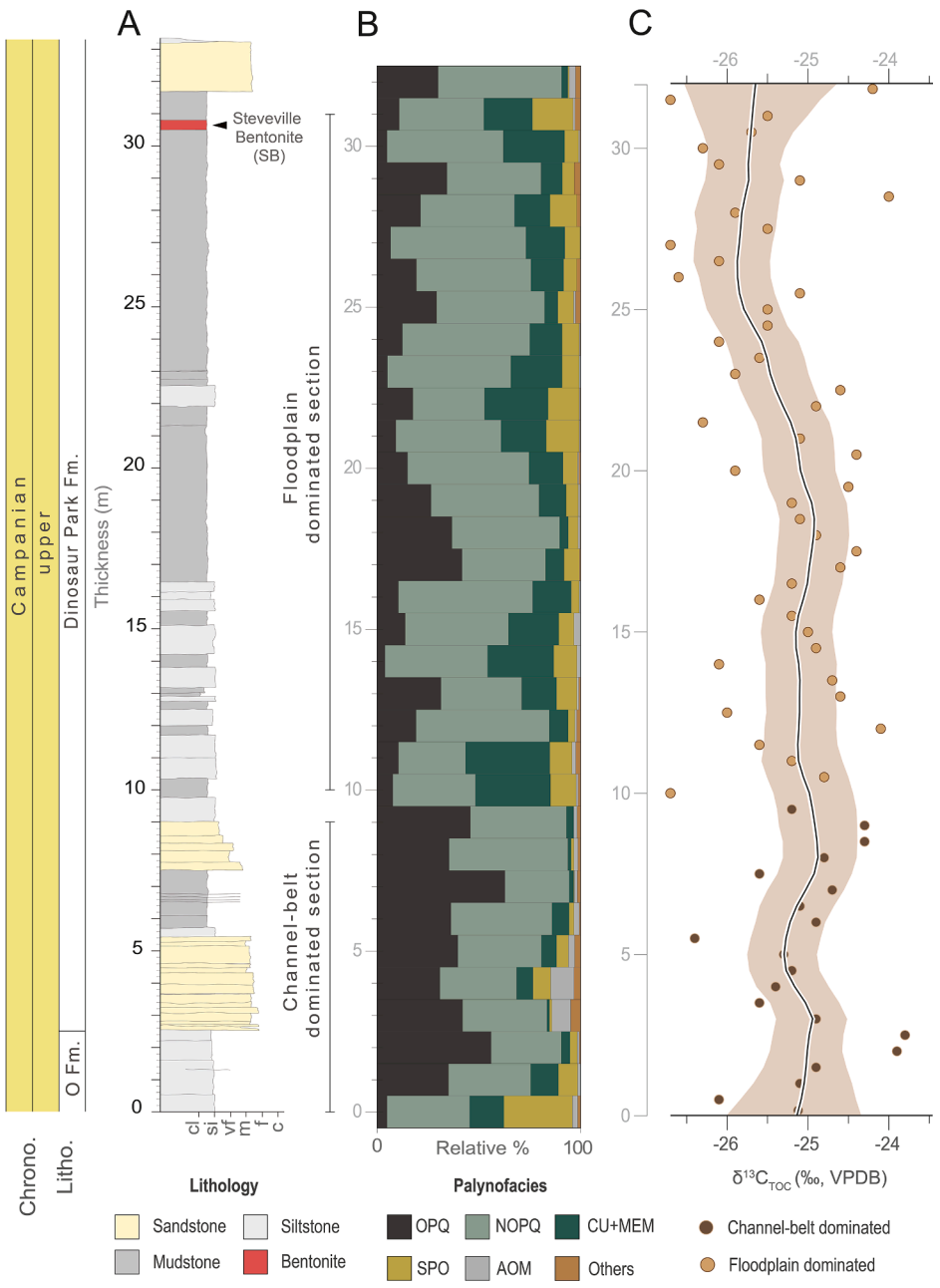


Fig. 2. (A) Sedimentary log of the measured section of the Dinosaur Park Formation in Steveville, Dinosaur Provincial Park (Alberta, Canada). cl - claystone; si - siltstone; vf - very fine sandstone; f - fine sandstone; m - medium sandstone; c - coarse sandstone. (B) Variation of kerogen assemblages along the measured section. OPQ - Opaque Phytoclasts; NOPQ - Non-opaque Phytoclasts; CU + MEM - Cuticles and Membranes; SPO - Sporomorphs; AOM - Amorphous Organic Matter. (A)–(B) modified from (Goonetilleke et al., 2025). (C) $\delta^{13}C_{TOC}$ record from the measured section fitted to a LOESS model (smoothing factor = 0.2, confidence interval = 95 %). Fm - Formation; O Fm – Oldman Formation; Chron - Chronostratigraphy; Litho - Lithostratigraphy. Lithostratigraphy from Eberth et al. (2023).

preserved in sedimentary successions (e.g., Suan et al., 2015; Silva and Alsen, 2024).

5.2.1. Channel-belt dominated section

The basal channel-dominated section is predominately composed of opaque (39 %) and non-opaque (36 %) phytoclasts (Fig. 2) (Goonetilleke et al., 2025). When all major kerogen assemblages from channel-dominated floodplains are plotted against $\delta^{13}C_{TOC}$, no significant correlations are determined (Fig. S3). The lack of correlation between individual kerogen groups and $\delta^{13}C_{TOC}$ suggests that changes in $\delta^{13}C_{TOC}$ cannot be attributed to variations in kerogen composition (e.g., cuticles vs opaque phytoclasts).

5.2.2. Floodplain dominated section

The upper stable floodplain section is predominantly composed of cuticles and membranes (20 %) and non-opaque phytoclasts (45 %) (Fig. 2) (Goonetilleke et al., 2025). Similar to the channel-dominated section, regression analysis of major kerogen assemblages from the floodplain shows no significant relationships (Fig. S4), suggesting that variations in preserved plant remnants cannot explain $\delta^{13}C_{TOC}$ trends observed within the stable floodplain interval.

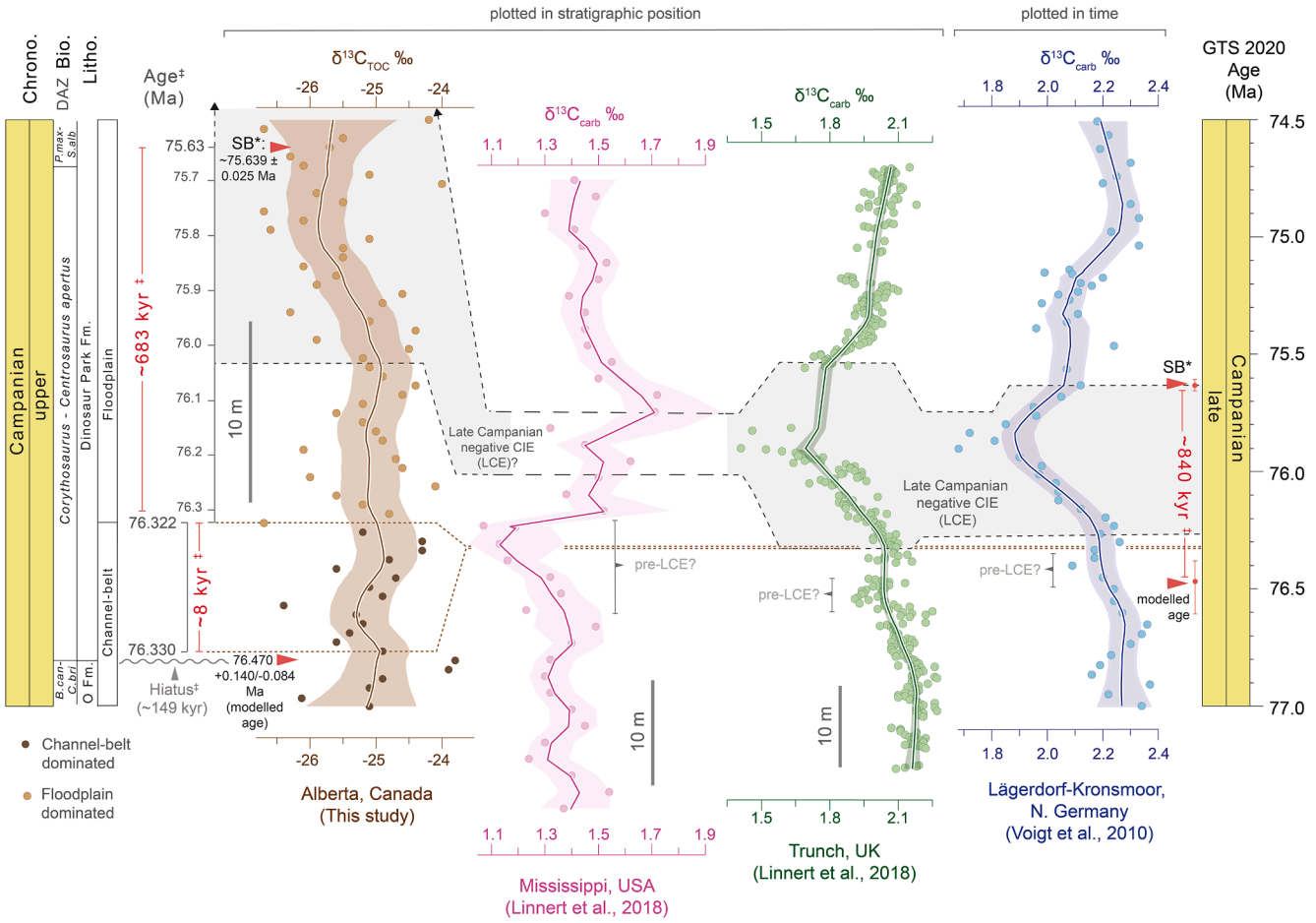


Fig. 3. Correlation of the Dinosaur Park Formation $\delta^{13}C_{TOC}$ record with another North American $\delta^{13}C_{carb}$ record and two European $\delta^{13}C_{carb}$ records from the Boreal Sea. All curves are fitted with a LOESS model (smoothing factor = 0.2 for Canada, the UK, and Germany; smoothing factor = 0.1 for the USA; confidence interval for all sections = 95 %). The section from Germany is calibrated to the GTS 2020. The radiometric and modeled ages for the Dinosaur Park Formation section are after Eberth et al. (2023) and Ramezani et al. (2022), which are correlated with the GTS 2020 (red arrow heads with their standard error). Age-calibrated biostratigraphy for the Dinosaur Park Formation section is after Eberth et al. (2023). The gray-shaded zone brackets the LCE. Black dashed lines with an arrow at the top of the chemozone in our section indicate that the LCE extends beyond the measured interval. SB*- Age of the Steveville Bentonite is correlated with that of the Plateau Tuff; Bio – Biostratigraphy; Chrono – Chronostratigraphy; Litho – Lithostratigraphy; DAB - Dinosaur assemblage biozones; *B. Can-C. Bri – Brachylophosaurus canadensis-Coronosaurus brinkmani* biozone; *P.max-S.alb - Prosaurolophus maximus- Styracosaurus albertensis* biozone. ‡ Refer to the age model (Fig. S2) and the supplementary text.

5.2.3. *Interpretation of the $\delta^{13}C_{TOC}$ record*

In alluvial deposits, the $\delta^{13}C$ signatures of bulk sedimentary organic compounds mainly originate from cuticles, membranes, and other decay-resistant plant residues (Tyson, 1995; Arens and Jahren, 2000). The studied section from the Dinosaur Park Formation is predominantly composed of continental kerogen assemblages (Fig. 2), suggesting that the $\delta^{13}C_{TOC}$ signal in this study can be attributed to preserved fossil remnants of autotrophs. Further, the secular variation observed within the Dinosaur Park Formation cannot be attributed to differences in metabolic processes in plants (i.e., C_3 vs C_4), given the lack of evidence for the evolution of C_4 physiology during the Mesozoic (Gröcke, 2002). Additionally, the $\delta^{13}C_{TOC}$ values from this study fluctuate near those associated with fossilized C_3 physiology (Gröcke, 2002; Silva et al., 2013), suggesting the dominance of C_3 vegetation throughout the measured section. Moreover, the absence of marine palynomorphs in the floodplain section (Goonetilleke et al., 2025), indicates no marine influence. Therefore, the carbon source from the preserved organic compounds is continental and primarily derived from photosynthesis, indicating that the obtained $\delta^{13}C_{TOC}$ reflects $\delta^{13}C$ of the contemporaneous atmospheric CO_2 (Gröcke, 2002). The well-preserved kerogen assemblages (Goonetilleke et al., 2025) and the low maturity coals of the overlying LCZ (Fig. 1A) (Beaton et al., 2006) indicate that the studied section has not undergone significant diagenesis that could substantially alter $\delta^{13}C_{TOC}$ values (Chung and Sackett, 1979; Lewan, 1983; Hoefs, 2015; Silva et al., 2017).

However, the $\delta^{13}C_{TOC}$ record from our section shows large fluctuations of up to ~3 ‰ (Fig. 2), especially when compared to the more stable marine records from the UK and N. Germany (Fig. 3). Marine $\delta^{13}C_{carb}$ records, from the Campanian and other Cretaceous time intervals, are often derived from relatively homogeneous carbonate sources such as calcareous nannofossils or planktonic foraminifera (e.g., Voigt et al., 2010; Linnert et al., 2018), resulting in less noisy $\delta^{13}C_{carb}$ trends. Yet, when compared with European sections, the Mississippi section shows a slightly elevated variability, suggesting a possible regional overprint on its $\delta^{13}C$ record (Fig. 3). The overall less noisy nature of $\delta^{13}C_{carb}$ records is also partly a reflection of the long residence time of the oceanic dissolved inorganic carbon (DIC) reservoir (Berner et al., 1999), effectively buffering short-term carbonate carbon isotope variability.

On the other hand, fluctuations of 2–3 ‰ have been reported from terrestrial environments during the Mesozoic (e.g., Arens and Jahren, 2000) and the Cenozoic (e.g., Baczynski et al., 2016). These variations are often attributed to natural terrestrial processes, such as seasonal changes in plant physiology (can fluctuate up to 1–2 ‰ in modern plants; Ehleringer et al., 1992), as well as changes in precipitation and temperature (Arens et al., 2000; Gröcke, 2002; Rodrigues et al., 2019). Given the strong fluctuating nature in terrestrial $\delta^{13}C$ records, epicontinental sections closer to continents, such as the Mississippi section from Linnert et al. (2018), are more likely to be affected by a terrestrial $\delta^{13}C$ overprint. This may explain why the Mississippi section exhibits slightly elevated variability compared to the contemporaneous European sections located farther offshore. Arens et al. (2000) demonstrate that, despite these effects, C_3 plant material, similar to the material preserved in our section, can still faithfully reflect the atmospheric $\delta^{13}C$ signal. Moreover, applying a LOESS model to the fluctuating $\delta^{13}C_{TOC}$ values allows extraction of a general trend from the noisy record, reflecting the broader atmospheric $\delta^{13}C$ changes over time.

In summary, the analysis of the $\delta^{13}C_{TOC}$ record from the Dinosaur Park Formation indicates that the observed broad secular trend does not correspond to major changes in the preserved organic matter, natural terrestrial processes, or diagenesis. Instead, we interpret these fluctuations to reflect the primary (global) signal, i.e., changes in the $\delta^{13}C$ of the atmospheric CO_2 (Gröcke, 2002).

5.3. Carbon-isotope stratigraphy of the Dinosaur Park Formation

The LCE corresponds to a negative CIE recorded globally, spanning regions from the Atlantic (Linnert et al., 2018) to the Pacific (Jung et al., 2013) oceans and from the Northern (Voigt et al., 2010) to the Southern Hemispheres (Thibault et al., 2012), with many of these records coming from the NW Tethys and the Boreal Sea (see Wendler, 2013). While all these records are confined to marine environments, they traverse from oceanic to shelf settings and capture the LCE at different latitudes with varying magnitudes, all below 1 ‰ (e.g., Jarvis et al., 2002; Voigt et al., 2010; Wendler, 2013; Linnert et al., 2018; Pearce et al., 2022). Changes in magnitude from shelf to oceanic setting have been discussed in Voigt et al. (2010 and 2012). These studies show that shelf environments generally exhibit higher magnitude C-excursions than deeper marine environments. It is also suggested that these differences arise either from how the two depositional settings respond differently to diagenesis or from variations in the partitioning of carbon cycling in each environment. The latitudinal impact on the magnitude of the LCE remains unclear. Tropical deep marine records (e.g., ODP site 1210B in Jung et al., 2012) do not show a prominent LCE signal, whereas mid-latitude sections, particularly on the European shelf, display a strong signal (Fig. 3). Such variations in the magnitude of CIEs with respect to latitude have been documented and discussed for other C-excursions in the geological record (e.g., van Bentum et al., 2012; Rodrigues et al., 2021). However, at this point, it is still unclear how latitudinal changes affect the magnitude of the LCE because: (1) comparisons between deep-marine and shelf records are incomplete (as in Voigt et al., 2010; 2012), and (2) there are no sufficient bulk $\delta^{13}C_{carb}$ records from the tropics for direct comparison with mid-latitudes.

The high-resolution $\delta^{13}C_{carb}$ curves from the shelves of Trunch (UK) and Lägerdorf-Kronsmoor (N. Germany) record the LCE negative CIE with a magnitude of ~0.6 ‰ and ~0.5 ‰, respectively (Fig. 3). With the LCE lasting around 450 kyr (Voigt et al., 2012), its inflection point is placed at ~75.9 Ma in the 2020 Geological Time Scale (Cramer and Jarvis, 2020; Gale et al., 2020) (Fig. 3). The biostratigraphic placement of the LCE has slight discrepancies (see Linnert et al., 2018) but is closely associated with the *Radotruncana calcarata* (planktonic foraminifer) biozone (Gale et al., 2020) (Fig. 3). Additionally, more recent studies (e.g., Chenot et al., 2016; Linnert et al., 2018) have identified another event before the LCE, informally referred to as the pre-LCE. The pre-LCE has a negative magnitude of ~0.2 ‰, which also falls within the *R. calcarata* biozone (see Chenot et al., 2016 and references therein). In Mississippi (USA), Linnert et al. (2018) identified the ~0.3 ‰ negative CIE as the pre-LCE, with the main event being even less prominent (Fig. 3). This suggests that the LCE is not strongly expressed across all regions (Fig. 3).

Given the secular nature of the LCE, our efforts to correlate the $\delta^{13}C$ stratigraphy of the Dinosaur Park Formation will be focused on the stable floodplain deposit. Although limitations exist in comparing $\delta^{13}C$ values between carbonate and organic systems (see Gröcke, 2020), the negative excursion of ~0.8 ‰ in the upper portion of the stable floodplain interval is here interpreted as the LCE negative CIE. Relatively larger magnitudes recorded in terrestrial environments compared to marine ones, as seen in our study, are common (e.g., Hesselbo et al., 2007) and have been attributed to changes in plant carbon fractionation in response to fluctuations in atmospheric CO_2 levels (Schubert and Jahren, 2013).

Our interpretation of the LCE record in the study section is supported by the radiometric ages from the Dinosaur Park Formation (Figs 2 and 3). As the Steveville Bentonite stratigraphically correlates to the Plateau Tuff from DPP, we can place the infection point of the negative CIE in the Dinosaur Park Formation to be older than 75.639 ± 0.025 (2σ internal uncertainties) Ma and younger than 76.470 + 0.140/-0.084 (2σ internal uncertainties) Ma (Fig. 3). The chronostratigraphic placement of the LCE in the Dinosaur Park Formation aligns with other sections documenting the event (Fig. 3). However, the full CIE is not likely captured in our measured section (Fig. 3), and we infer that it would have been preserved in the overlying floodplain strata if the channel had not avulsed back to our section location (Fig. 2), eroding that portion. Moreover, events with small magnitudes, such as the pre-LCE (if present within the measured interval), may not be captured by the LOESS model due to the highly noisy nature of the terrestrial $\delta^{13}C_{TOC}$ record (Fig. 3). Regardless, the presence of the initial LCE negative CIE and inflection point to background values within our studied section supports our hypothesis that the Dinosaur Park Formation captures global-scale carbon cycling changes and that the Dinosaur Park Formation $\delta^{13}C$ terrestrial record can serve as chemostratigraphic tools for correlating with sedimentary sections worldwide.

5.4. What are the implications of recognizing the LCE in the Dinosaur Park Formation?

Unlike most negative CIEs, the LCE has been linked to a cooling event, as indicated by oxygen-isotope and TEX_{86} records (Linnert et al., 2014, 2018). Linnert et al. (2018) suggested that the LCE was caused by a series of long-lasting events linked to tectonic (e.g., changes in seafloor spreading rates) and paleoceanographic (e.g., changes in global ocean circulation) changes, which led to a decrease in atmospheric CO_2 and subsequent climate cooling. Chenot et al. (2016 and 2018) added that during the LCE, weathering and erosion were high in terrestrial environments in Europe, implying the reworking and oxidation of organic matter to CO_2. The cooling trend towards the late Campanian, documented in British Columbia (Canada) (Zakharov et al., 2013), along with the increasing terrigenous input trend starting around the middle

Campanian in the Northwestern Territories (Canada) (Diaz et al., 2025), suggests a similar driving force for the LCE within Canada.

The Dinosaur Park Formation is commonly divided into at least two distinct dinosaur faunal assemblage zones, with the lower part characterized by the abundance of *Corythosaurus* and *Centrosaurus*, replaced upwards by *Prosaurolophus* and *Styracosaurus* (Mallon et al., 2012). Our study demonstrates that this transition occurs at, or near, the LCE negative CIE (Fig 3). In addition to the Dinosaur Park Formation 's globally renowned dinosaur fossil record, featuring 50 dinosaur taxa, there are more than 450 other fossilized flora and fauna, making the Dinosaur Park Formation sedimentological record the best snapshot of a continental Late Cretaceous ecosystem to date (Currie and Koppelhus, 2005; Mallon et al., 2012; Brown et al., 2013; Cullen and Evans, 2016). Acknowledging that a portion of the Dinosaur Park Formation was deposited during the LCE and establishing a precise correlation with other global sections provides critical new insights into the evolution of dinosaurs and their associated ecosystems during periods of global carbon cycle disruptions, as well as the potential geodynamic and environmental drivers behind these changes. Furthermore, placing the Dinosaur Park Formation within the context of the LCE allows for a re-evaluation of previous studies on the evolution of dinosaurs and other contemporary groups, framed by the known planetary-scale Earth system disturbances of the Late Cretaceous.

6. Conclusions

The $\delta^{13}C$ record from a terrestrial floodplain-dominated section of the Upper Cretaceous Dinosaur Park Formation in Dinosaur Provincial Park, Alberta, Canada, leads to the following conclusions:

1. The primary $\delta^{13}C_{TOC}$ signal preserved in the floodplains of Dinosaur Park Formation reflects the global $\delta^{13}C$ signal, i.e., changes in the $\delta^{13}C$ of the atmosphere.
2. Part of the Dinosaur Park Formation was deposited during the Late Campanian Event (LCE).
3. Campanian-aged stable floodplain terrestrial records can serve as chemostratigraphic tools for correlating sedimentary sections worldwide.

CRediT authorship contribution statement

Muditha Goonetilleke: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ricardo L. Silva:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Paul R. Durkin:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Muditha Goonetilleke reports financial support was provided by Natural Sciences and Engineering Research Council of Canada. Muditha Goonetilleke reports a relationship with Natural Sciences and Engineering Research Council of Canada that includes: funding grants. Paul R. Durkin reports a relationship with Natural Sciences and Engineering Research Council of Canada that includes: funding grants. Ricardo L. Silva reports a relationship with Natural Sciences and Engineering Research Council of Canada that includes: funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Supplementary data to this article can be found online at.

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Appendix A. Supplementary data

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